Lab Test Kit Introduction and Basic Logic Functions

Lab 1



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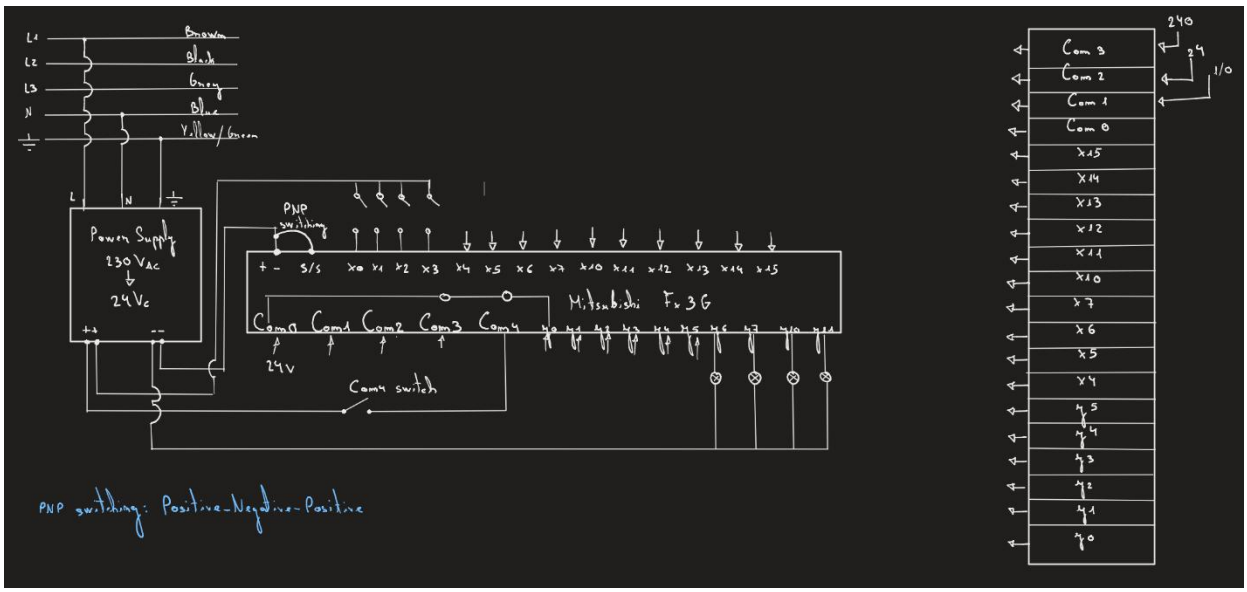
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Section 1 – Mitsubishi Test Kit Analysis

1. Mitsubishi FX3G PLC Sketch

Sketch the complete Mitsubishi FX3G PLC training kit on a full A4 sheet. Include the physical PLC, 24 V DC power supply, toggle switches, input and output devices, HMI, terminal blocks and all associated wiring. Clearly identify the PLC inputs (X), outputs (Y), power supply connections, 0 V/common connections and HMI communication connection. Produce a complete wiring schematic showing how all components of the training kit are interconnected.



2. Define the difference between contacts

Define the difference between a normally open and normally closed contact and give a brief example where each could be used to perform a logical operation.

A normally open contact is one that, in its resting state, does not allow current to pass to the circuit. It only allows current to flow when activated. A practical example is a Start button on a machine's control panel.

A normally closed contact is exactly the opposite. In its resting state, it allows current to flow, whereas upon activation, it stops current from passing to the rest of the circuit. An example is an emergency stop button.

3. Troubleshooting potential electrical fault

State clearly when troubleshooting potential electrical fault on the test rig using your digital multi-meter, state the very first step that should take place on the test equipment and why.

When tracking down an electrical issue on the test rig with a digital multimeter, the very first thing you need to check is the power supply. You need to check that it is

getting the 230V AC from the mains and providing properly the required DC voltages for the system.

If it fails or is turned off, the entire PLC loses power. Doing this step first is key to ruling out a basic power drop before moving on to trickier diagnostics with the wiring or the ladder logic.

Section 2 – Truth Table Analysis

1. Construct a ladder diagram for each of the circuits shown below. Create a new diagram for each circuit.
2. Test the ladder logic to ensure there are no errors or warnings.

Circuit a:

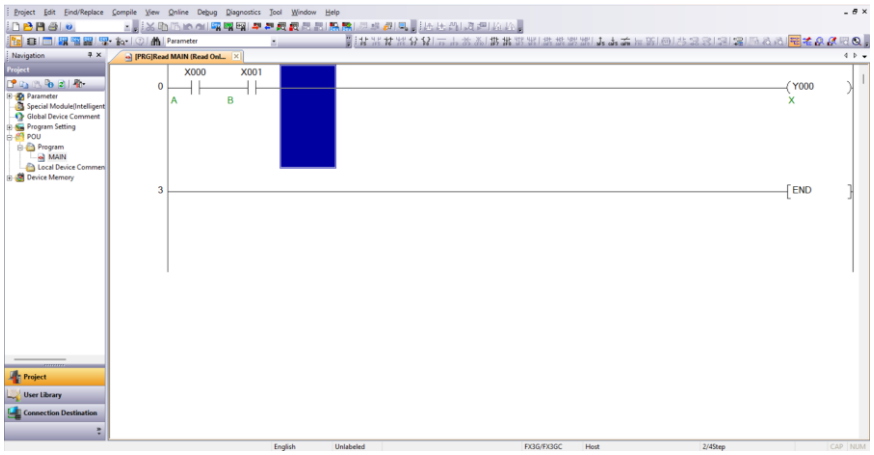


Figure 1 Circuit a

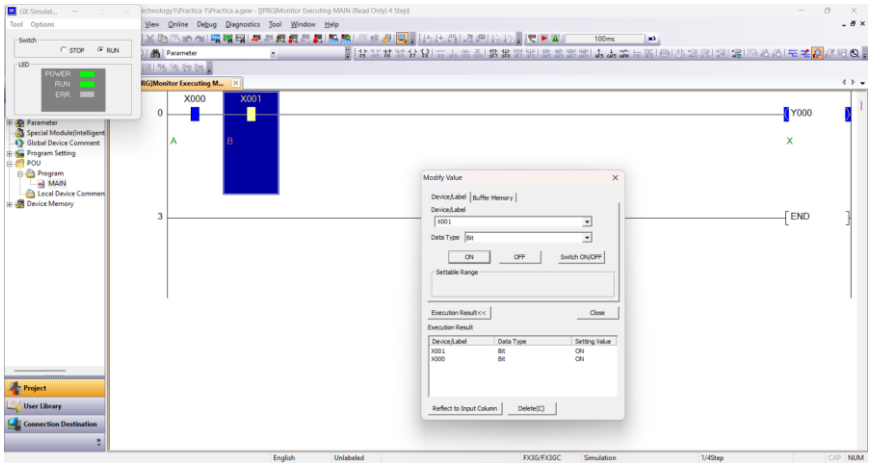


Figure 2 Test a

Circuit b:

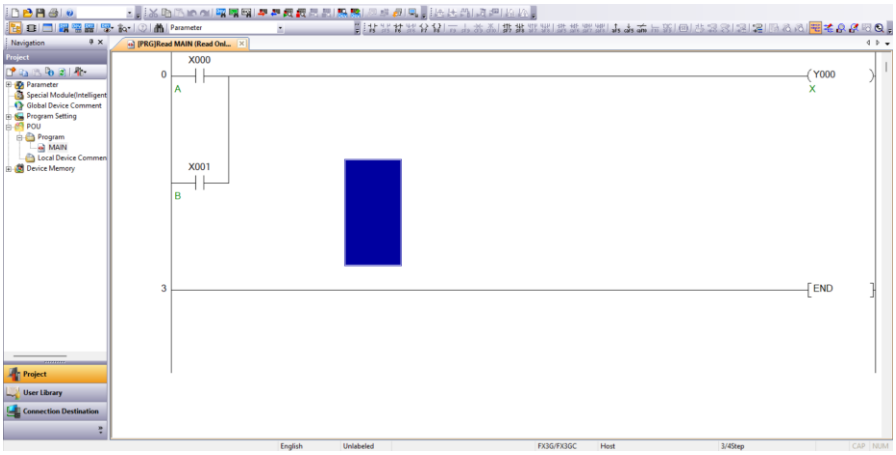


Figure 3 Circuit b

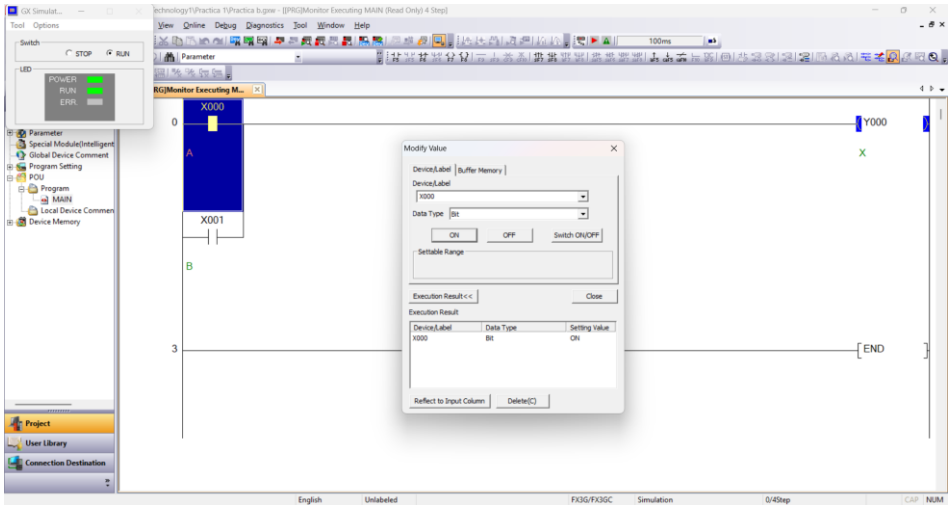


Figure 4 Test b

Circuit c:

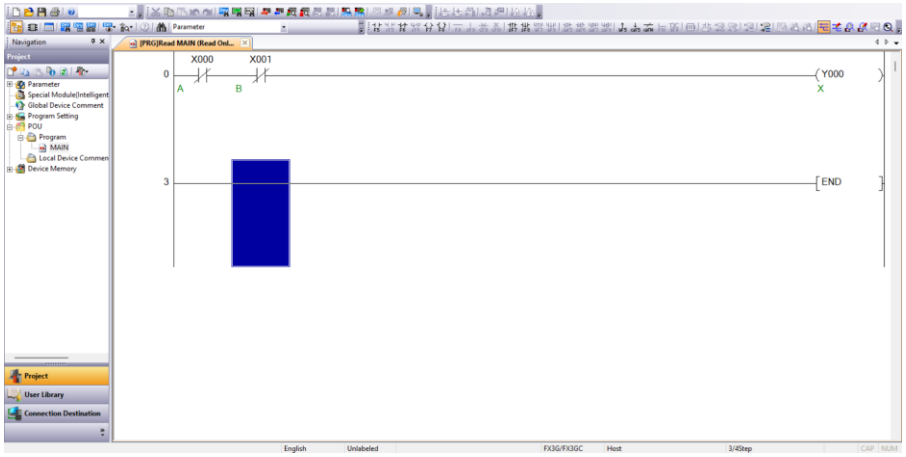


Figure 5 Circuit c

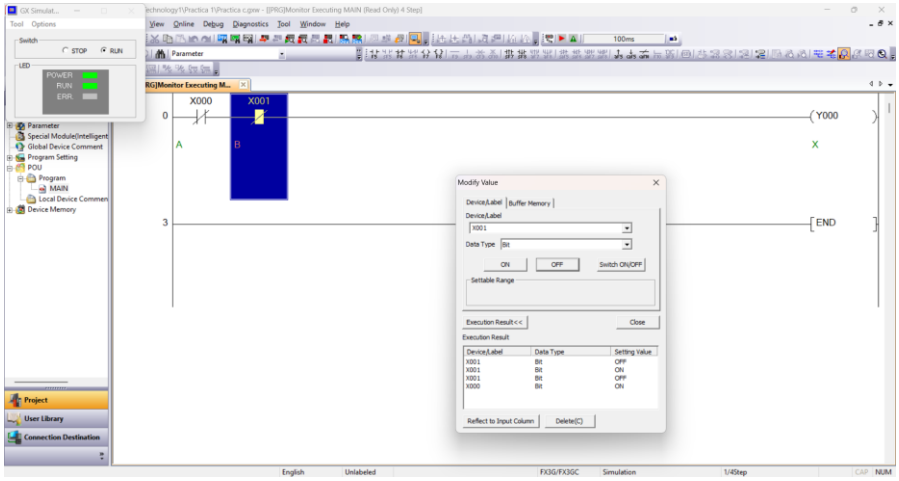


Figure 6 Test c

Circuit d:

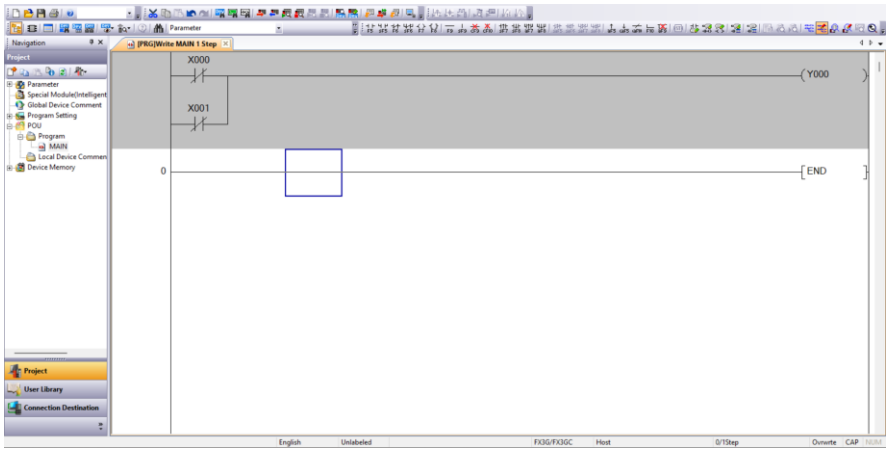


Figure 7 Circuit d

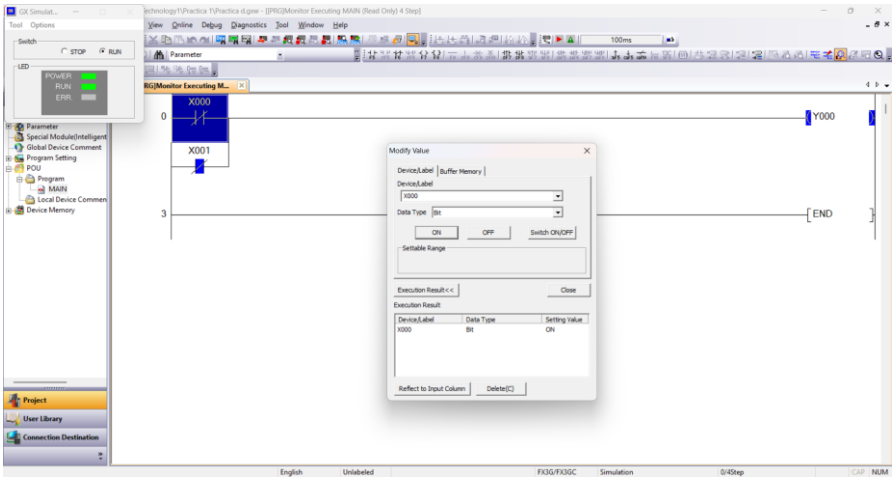


Figure 8 Test d

Circuit e:

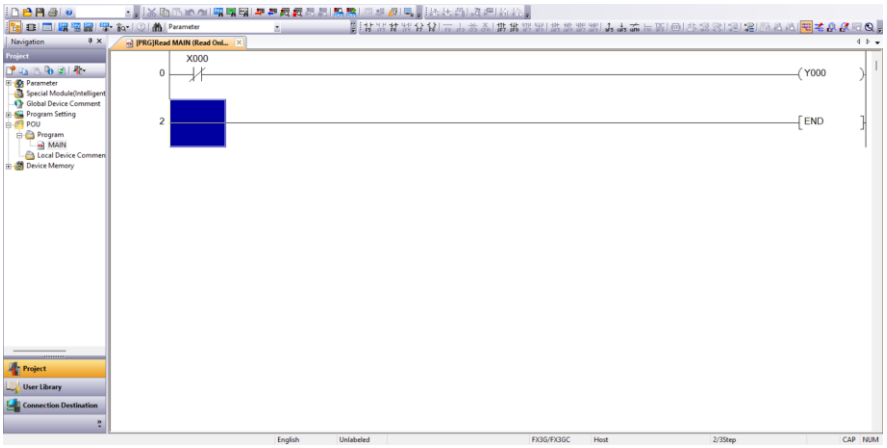


Figure 9 Circuit e

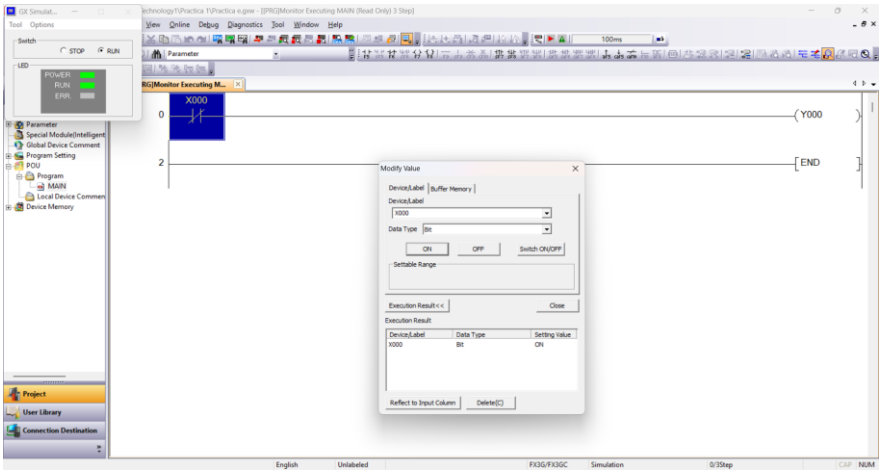


Figure 10 Test e

Circuit f:

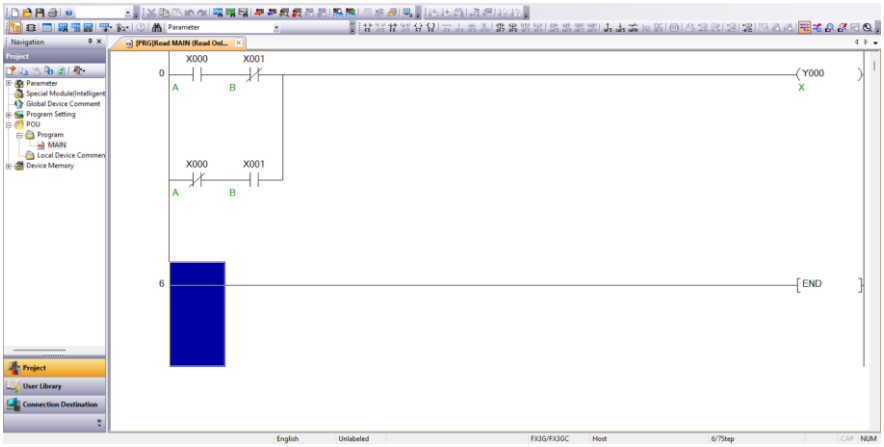


Figure 11 Circuit f

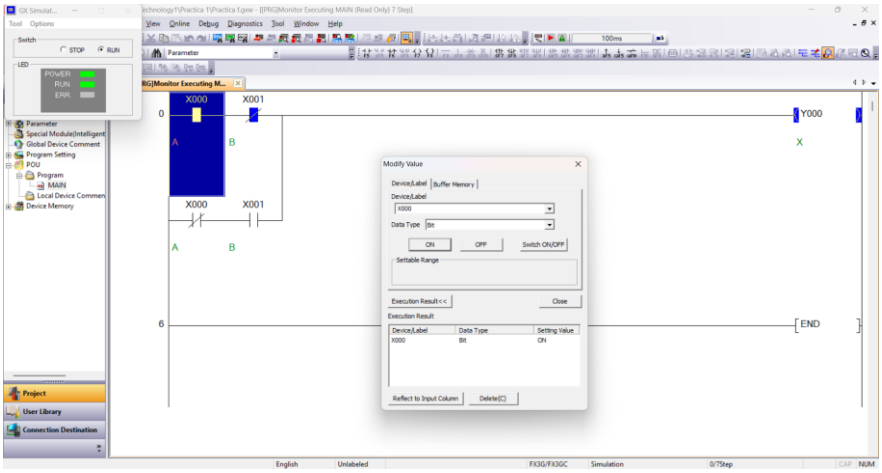


Figure 12 Test f

3. Using truth tables, establish the logic function for each circuit. Use Excel to generate a table for all variables.

See excel document attached.

	Circuit a)	AND
A	B	X
0	0	0
0	1	0
1	0	0
1	1	1

	Circuit b)	OR
A	B	X
0	0	0
0	1	1
1	0	1
1	1	1

	Circuit c)	NOR
A	B	X
0	0	1
0	1	0
1	0	0
1	1	0

	Circuit d)	NAND
A	B	X
0	0	1
0	1	1
1	0	1
1	1	0

Circuit e)	NOT
A	X
0	1
1	0

	Circuit f)	XOR
A	B	X
0	0	0
0	1	1
1	0	1
1	1	0

Figure 13 Truth tables

Section 3

1. Diagram

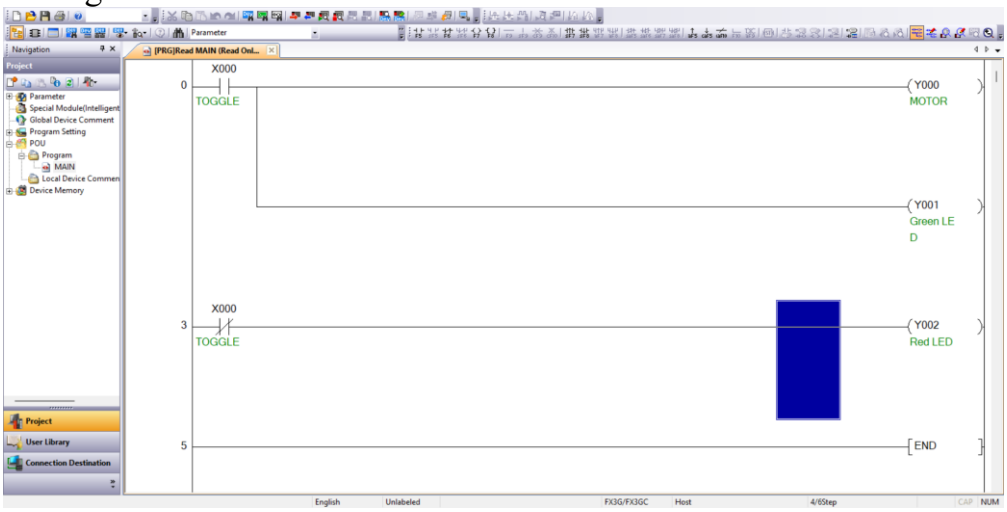


Figure 14 Diagram complete

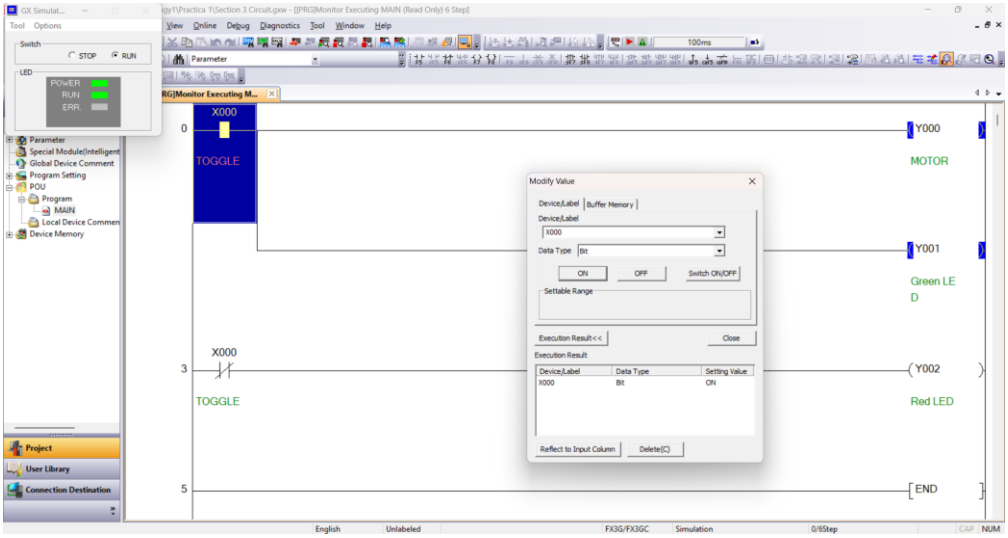
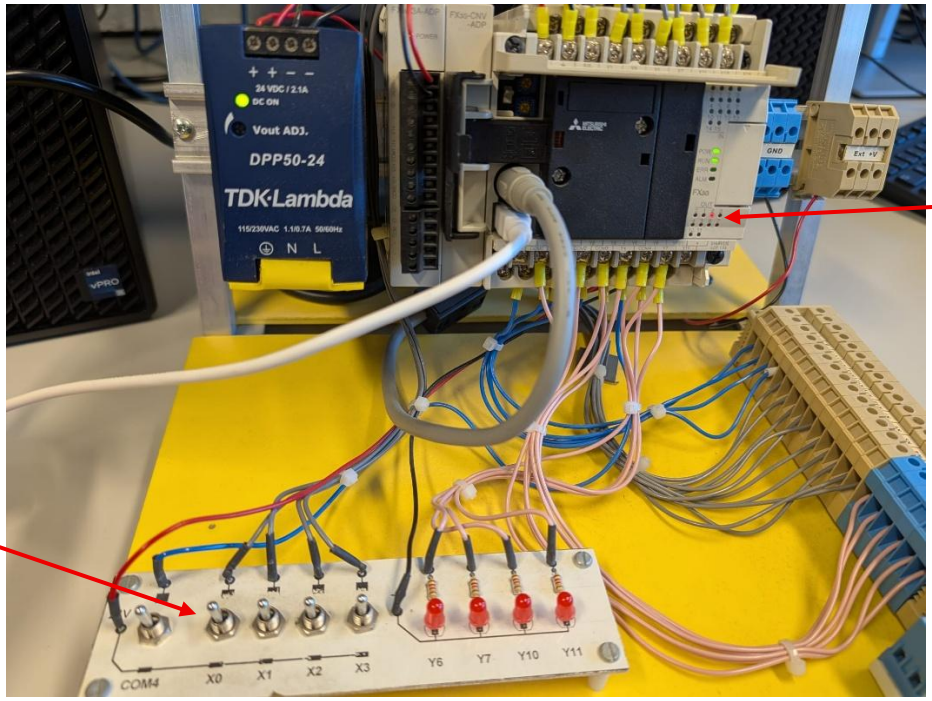


Figure 15 Diagram Test

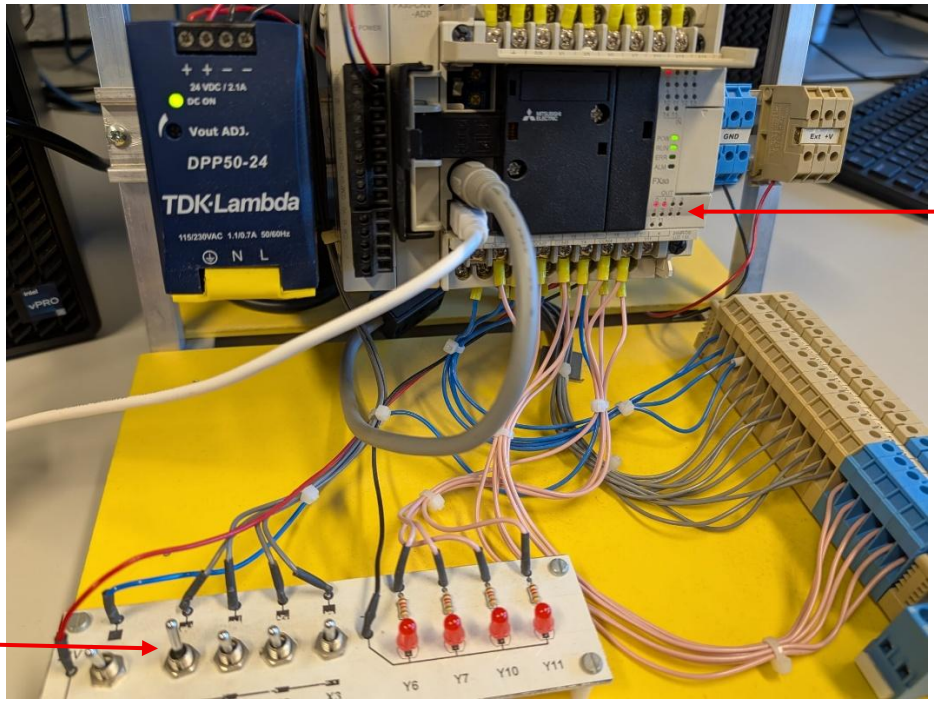
2. PLC test unit



Y0 (motor) is off and Y2 (red LED) is on

X0 switch is low

Figure 16 Circuit with X0 low



Y0 and Y1 are on, motor and green led respectively

X0 switch is high

Figure 17 Circuit with X0 high

3. Three-phase (415V) supply Diagram

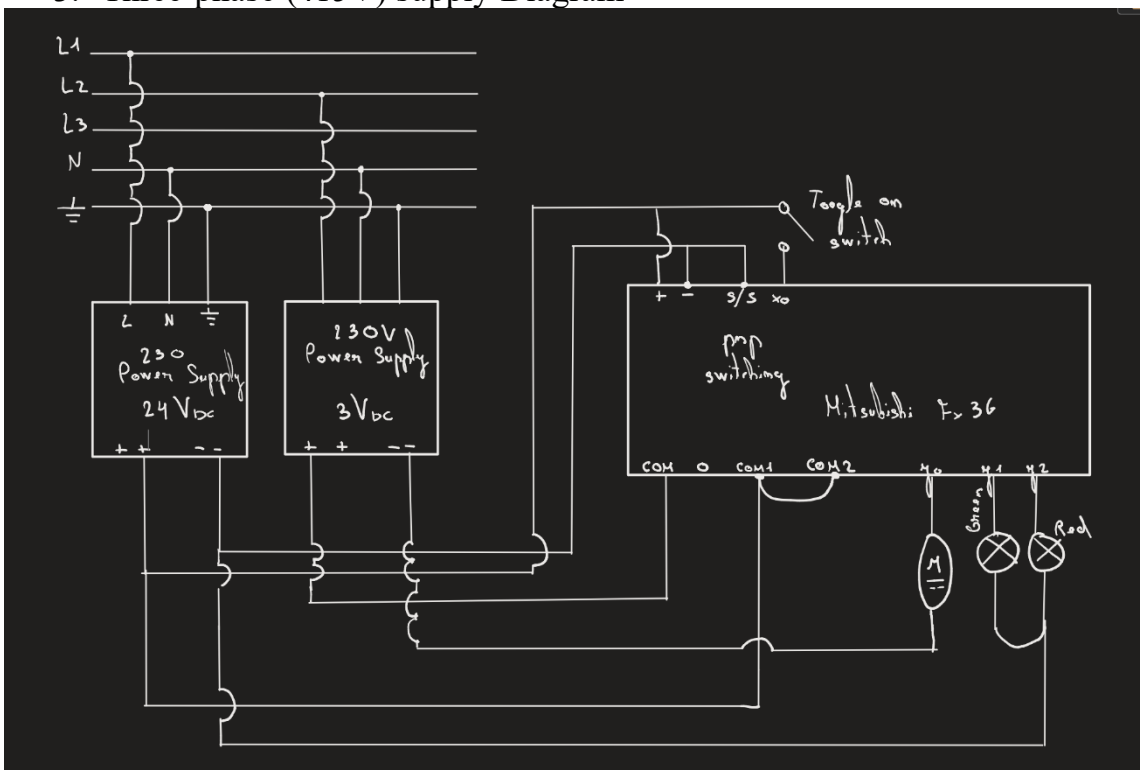


Figure 18 Supply power to the PLC and auxiliaries from a three-phase (415V) supply Diagram

4. Conclusion and Discussion of results

During this lab practical, an introduction was provided to the Mitsubishi FX3G PLC training kit, the various types of contacts, the use of basic logic operations, Ladder diagram programming, and the GX Works2 software.

It also covered the correct execution of programs and debugging tests within GX Works2 for the different Ladder programming exercises. This approach made it possible to demonstrate that these exercises corresponded to the AND, OR, NOR, NAND, NOT, and XOR logic operations.

Finally, the last part of the assignment helped to bridge this theoretical knowledge with a preliminary industrial application through the implementation of a motor and control LEDs.